

Chapter-2

2.1. a) Closure: Both addition and multiplication are closed on the set of real numbers, meaning that the sum or product of any two real numbers is also a real number.

b) Associative: Addition and multiplication are both associative, meaning that $(a+b)+c = a+(b+c)$ & $(ab)c = a(bc)$ for all real numbers a, b & c .

c) commutative: Addition & multiplication are both commutative, meaning that $(a+b) = b+a$ & $ab = ba$ for all real numbers a, b & c .

d) Identity: Addition has an identity element of 0, meaning that $a+0 = a$

for all real numbers a , multiplication has an identity element of 1, meaning that $a \times 1 = a$ for all real number a

e) Inverse: Addition has an inverse element $-a$ for each real number a , meaning that $a + (-a) = 0$. Multiplication has an inverse element of $\frac{1}{a}$ for each non-zero real number a , meaning that $a \times (\frac{1}{a}) = 1$

f) Distributive: multiplication distributes over addition meaning that $a(b+c) = ab+ac$ for all real numbers a, b, c

2.2. The huntington postulates state that a boolean algebra must satisfy the following conditions for all elements x, y, z in the set:

i) $x + y = y + x$

$$ii) x + (y + z) = (x + y) + z$$

$$iii) x + x = x$$

all postulates are satisfied except for postulate 7; there is no complement

2.3) a) The associative laws:

$$x(yz) = (xy)z$$

x	y	z	yz	x(yz)	xy	(xy)z
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	1	0	0	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	0	1	0
1	1	1	1	1	1	1

b) De-Morgan's Theorem :

$$(x+y+z)' = x' y' z'$$

x	y	z	x'	y'	z'	x'y'z'	x+y+z	(x+y+z)'
0	0	0	1	1	1	1	0	1
0	0	1	1	1	0	0	1	0
0	1	0	1	0	1	0	1	0
0	1	1	1	0	0	0	1	0
1	0	0	0	1	1	0	1	0
1	0	1	0	1	0	0	1	0
1	1	0	0	0	1	0	1	0
1	1	1	0	0	0	0	1	0

c) The distributive law:

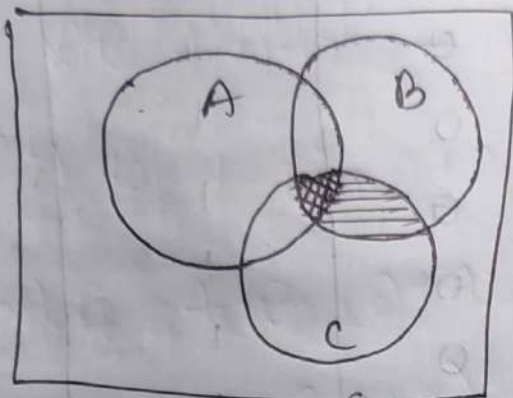
$$x + (y \cdot z) = (x + y) \cdot (x + z)$$

x	y	z	yz	(x+y)	(x+z)	(x+y)(x+z)	x+(yz)
0	0	0	0	0	0	0	0
0	0	1	0	0	1	0	0
0	1	0	0	1	0	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

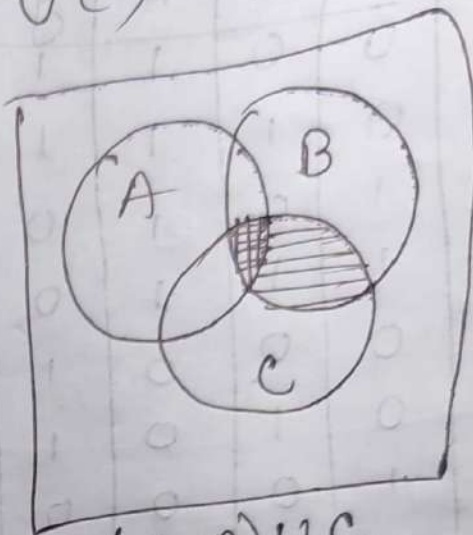
2.4) Associative laws

$$A(BC) = (AB)C$$

$$A \cup (B \cap C) = (A \cup B) \cap C$$



$A \cup (B \cap C)$

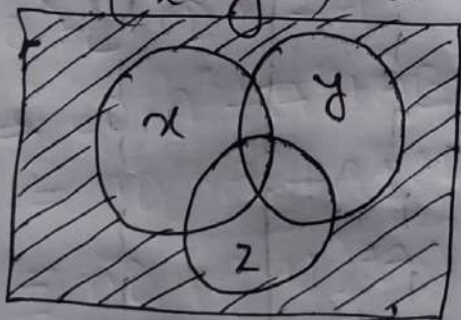


$(A \cup B) \cap C$

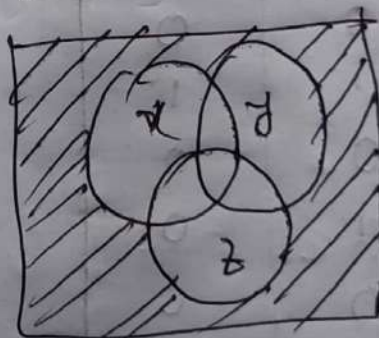
De Morgan's theorem for three variables

$$(x + y + z)' = x' y' z'$$

$$(x y z)' = x' + y' + z'$$



$x y z'$

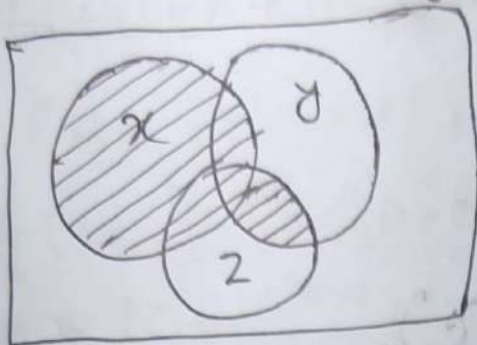


$x' y' z'$

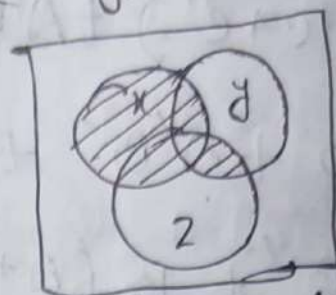
Distributive law:

$$x + (y \cdot z) = (x + y) \cdot (x + z)$$

$$x \cdot (y + z) = (x \cdot y) + (x \cdot z)$$



$$x \cup (y \cap z)$$



$$(x \cap y) \cup (x \cap z)$$

$$\begin{aligned} a) & x + x' \\ &= x + 1 \\ &= x \cdot 1 \\ &= x \end{aligned}$$

$$\begin{aligned} b) & (x + y) \cdot (x + y') \\ &= x \cdot x + x \cdot y' + y \cdot x + y \cdot y' \\ &= x + x \cdot y' + y \cdot x + 0 \\ &= x + x \cdot y' + y \cdot x \\ &= (x + y' + y) \cdot x \\ &= (x + 1) \cdot x \\ &= 1 \cdot x = x \end{aligned}$$

$$\begin{aligned} c) & x \cdot y + x \cdot z \\ &= x \cdot (y + z) \\ &= x \cdot (y + z) \end{aligned}$$

$$\begin{aligned} d) & x \cdot z + x' \cdot y \\ &= z \cdot (x + x') \end{aligned}$$

$$\begin{aligned} e) & (A + B) \cdot (A' + B') \\ &= (A' \cdot B') \cdot (A \cdot B) \\ &= 0 \end{aligned}$$

$$\begin{aligned} f) & x \cdot (y + z) + x' \cdot y \\ &= x \cdot y + x \cdot z + x' \cdot y \\ &= (x + x') \cdot y + x \cdot z \\ &= 1 \cdot y + x \cdot z \\ &= y + x \cdot z \end{aligned}$$

2.6) a) $ABC + A'B'C + A'BC + ABC' + A'B'C'$ to five literals

$$= AB(C+C') + A'B'(C+C') + A'BC$$

$$= AB + A'B' + A'BC$$

$$= A'B' + B(A+A'C)$$

$$= A'B' + B(A+A')(A+C)$$

$$= A'B' + B(A+C)$$

b) $BC + AC' + AB + BCD$

$$= BC(1+D) + A(B+C')$$

$$= BC + AB + AC'$$

$$BC = A(B+C) = BC + A(B+C)$$

c) $[(CD)' + A]' + A + CD + AB$

$$= CD \cdot A' + A + AB + CD$$

$$= CD \cdot A' + A + CD$$

$$= CD(A'+1) + A$$

$$= A + CD$$

d) $(A+C+D)(A+C+D')(A+C'+D)(A+B'D)$

$$= (A \cdot A + AC + AD' + AC + C \cdot C + CD' + AD + CDAD')$$

$$(A \cdot A + AB' + AC' + B'C' + AD + B'D)$$

$$= (A + AC + AD' + C + CD' + AD + CD + 0)(A + AB' + AC' + B'C' + AD + B'D)$$

$$= (A+C+AC+1+1) (A+AB'+Ac'+B'c'+AD+B'D)$$

$$= (1+1+AC) (A+AB'+Ac'+B'c'+AD+B'D)$$

$$= (1+1) (A+AB'+Ac'+B'c'+AD+B'D)$$

$$= A+AB'+Ac'+B'c'+AD+B'D$$

$$= A(1+B') + A(D+c') + B'(D+c')$$

$$= A + (D+c')(A+B')$$

$$2.7) a) (Bc'+A'D)(AB'+CD)'$$

$$= (Bc'+A'D)' + (AB'+CD)'$$

$$= (Bc')'(A'D)' + (AB')'(CD)'$$

$$= B'c \cdot AD' + A'Bc'D$$

$$= AB'cD' + A'Bc'D$$

$$= 1$$

$$b) B'D + A'BC + ACD + A'BC$$

$$= B'D + A'B + ACD$$

$$\text{So } (B'D + A'B + ACD)'$$

$$= (B'D + A'B)' \cdot (ACD)'$$

$$= (B'D)'(A'B)' \cdot (ACD)'$$

$$= (B+D')(A+B')(A'+(CD)')$$

$$= (AB+AD'+B'D')(A'+c'+D')$$

$$= ABc' + ABD' + AC'D' + AD' + A'B'D' + B'c'D' + B'D$$

$$= ABc' + BAD' + A'Ac' + DA + B'D$$

$$= A'BC' + AD' + B'D'$$

$$= B'D' + A(D' + BC')$$

$$c) [(AB)'A] [(AB)'B]$$

$$= ([(AB)'A] + [(AB)'B])$$

$$= [(AB)'A]' + [(AB)'B]'$$

$$= (AB + A') + (AB + B')$$

$$= AB + A' + AB + B'$$

$$= AB + A' + B'$$

$$= A' + AB + B'$$

$$= (A + A') (A' + B) + B$$

$$= A' + B + B'$$

$$= A' + 1$$

$$= 1$$

$$d) (AB' + C'D')$$

$$= (AB')' \cdot (C'D')'$$

$$= (A' + B)(C + D)$$

$$= A'C + A'D + BC + BD$$

$$2.8) F_1(A, B, C) = \sum m_i$$

$$F_2(A, B, C) = \sum m_i$$

$$a) E = F_1 + F_2 = \sum m_i + \sum m_i$$

$$= \sum (m_i + m_i)$$

$$b) G = F_1 F_2 = \sum m_i \sum m_i$$

Let assume,

$$F_1(A, B, C) = A'BC' + ABC + A'BC = \sum (2, 3, 7)$$

$$F_2(A, B, C) = ABC' + A'BC + A'B'C' = \sum (0, 3, 6)$$

$$E = F_1 + F_2 = A'BC' + ABC + A'BC + ABC' + A'BC + A'B'C'$$

$$= \sum (0, 2, 3, 6, 7)$$

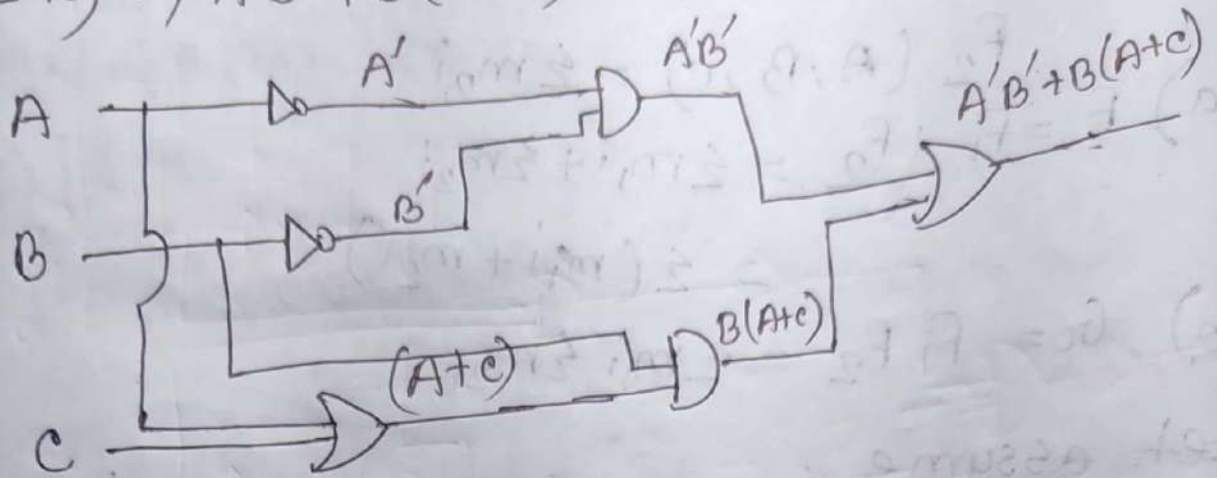
$$b) G = F_1 F_2 = (A'BC' + ABC + A'BC)(ABC' + A'BC + A'B'C')$$

$$= A'BC$$

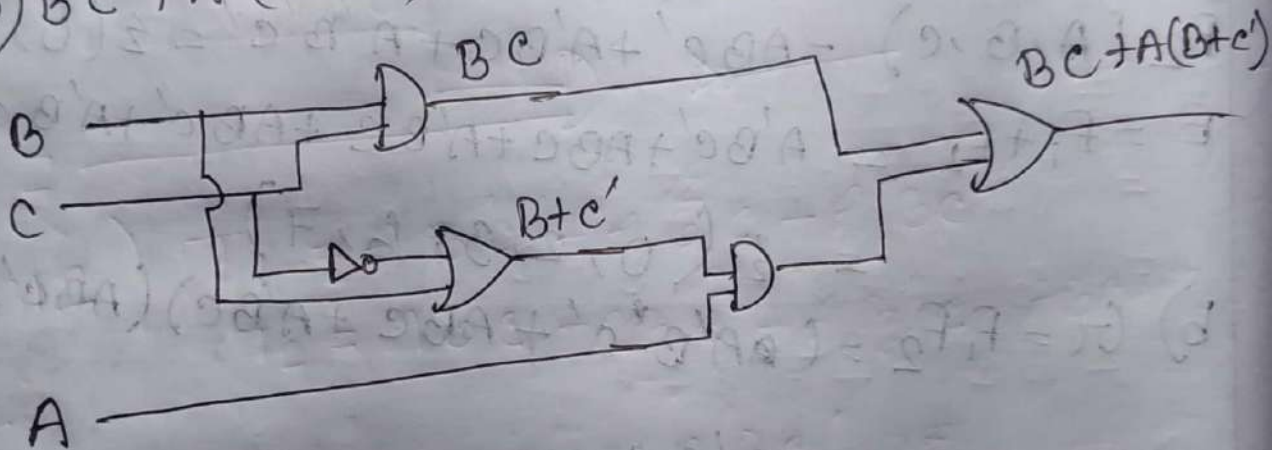
$$2.9) F = xy + \bar{x}y' + y'z$$

x	y	z	xy	y'	xy'	y'z	F = xy + \bar{x}y' + y'z
0	0	0	0	1	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	0
1	0	0	0	1	1	0	1
1	0	1	0	1	1	0	1
1	1	0	1	0	0	0	1
1	1	1	1	0	0	0	1

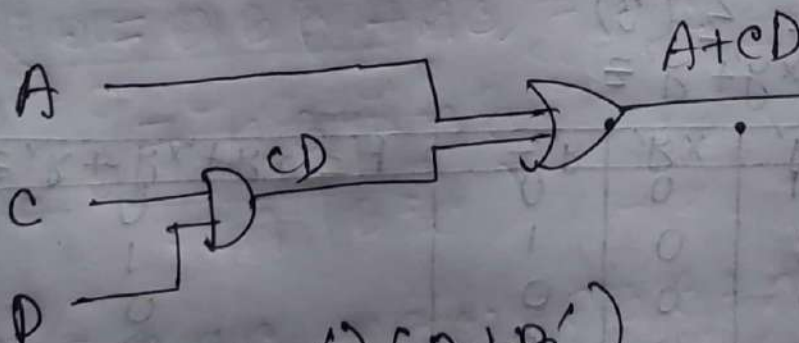
2.10) a) $A'B' + B(A+C)$



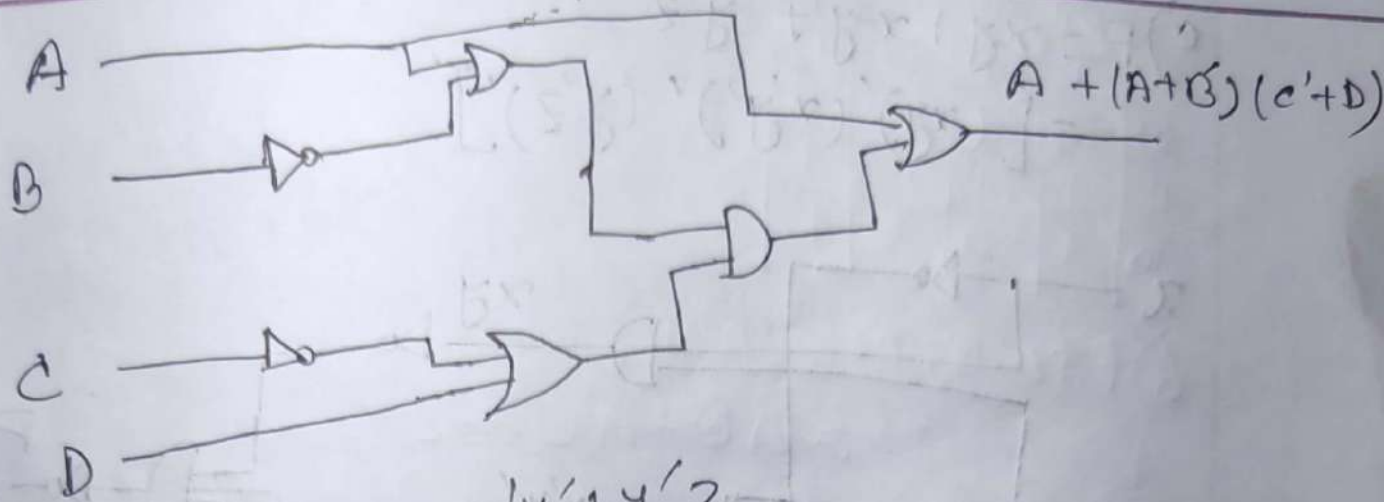
b) $BC + A(B+C')$



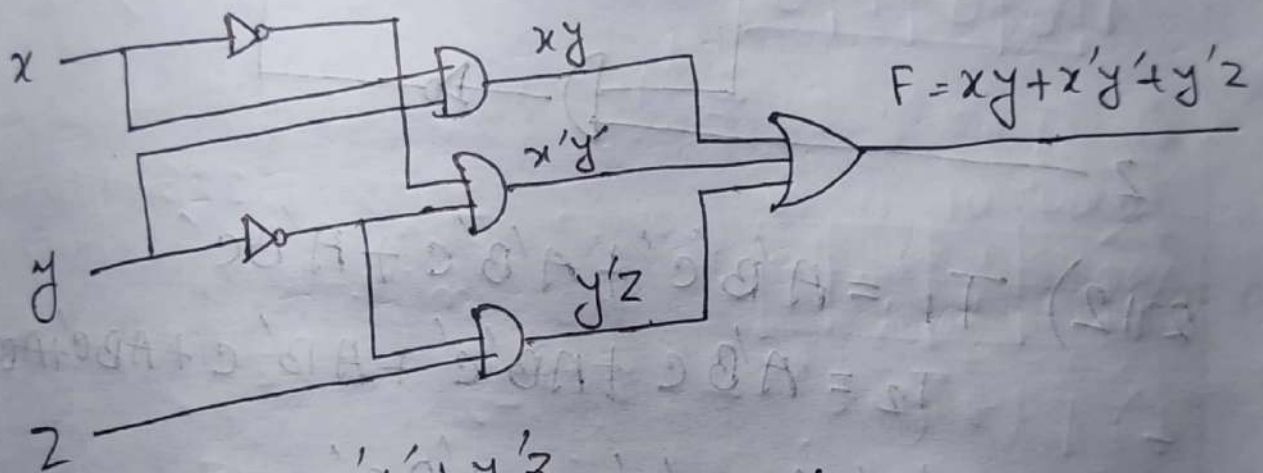
c) $A + CD$



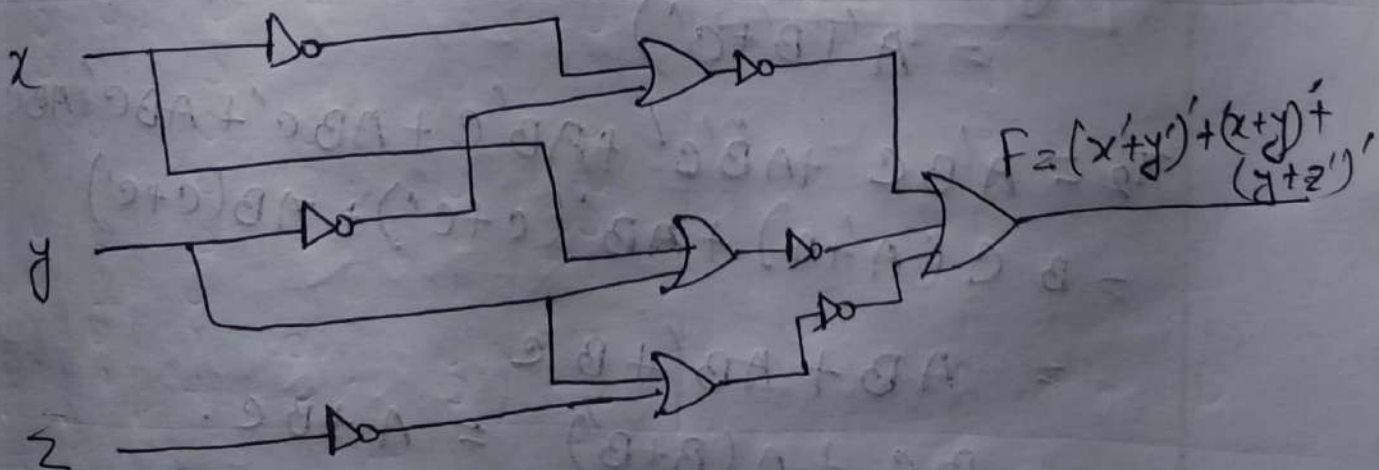
d) $A + (D+C')(A+B')$
 $= A + (A+B')(C'+D)$



2.11) a) $F = xy + x'y' + y'z$

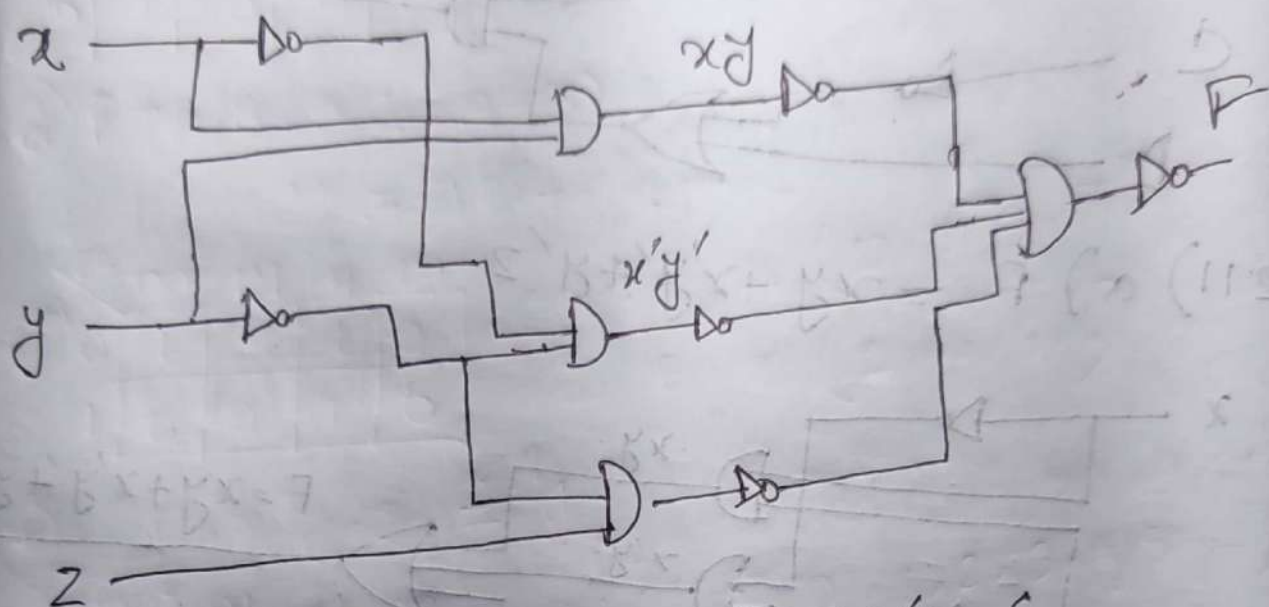


b) $F = xy + x'y' + y'z$
 $z(xy)' + (x'y')' + (y'z)'$
 $= (x'+y') + (x+y) + (z+y)'$



$$c) F = xy + x'y' + y'z$$

$$= [(xy)'(x'y')'(y'z)']'$$



$$2.12) T_1 = A'B'C' + A'B'C + A'BC'$$

$$T_2 = A'BC + AB'C' + AB'C + ABC' + ABC$$

$$T_1 = A'B'C' + A'B'C + A'BC'$$

$$= A'B'(C' + C) + A'C'(B + B')$$

$$= A'B' + A'C'$$

$$= A'(B' + C')$$

$$T_2 = A'BC + A'B'C' + AB'C' + AB'C + ABC' + ABC$$

$$= BC(A' + A) + AB'(C' + C) + AB(C' + C)$$

$$= BC + AB' + ABC$$

$$= BC + A(B + B') = A + BC$$

2.13) a) $F(A, B, C, D) = D(A' + B) + B'D$
 $= A'D + BD + B'D$

CD \ AB	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

SOP: $f = \sum(1, 3, 5, 7, 9, 11, 13, 15)$

POS: $F = \prod(0, 2, 4, 6, 8, 10, 12, 14)$

b) $F(w, x, y, z) = y'z + wx'y' + wxz' + w'x'z$

yz \ wx	00	01	11	10
00	0	1	1	0
01	0	1	0	0
11	1	1	0	1
10	0	1	0	0

SOP: $f = \sum(1, 3, 5, 9, 12, 13, 14)$

POS: $F = \prod(0, 2, 4, 6, 7, 8, 10, 11, 15)$

c) $f(A, B, C, D) = (A + B' + C)(A + B')(A + C' + D')$
 $(A' + B + C + D')(B + C' + D')$

$= (A + AB' + AB' + BC')(A + C' + D')(A'B + A'C' + A'D' +$

$B + BC' + BD' + BC + CD' + BD' + D'C' + D')$

$= (A + AB' + B'C')(A + C' + D')(A'B + A'C' + A'D' +$
 $B + BD' + BC + D' + D')$

$= (A + AB'C' + AB'D' + AB'C + BCD')(A'B + A'C' + A'D'$
 $+ B + D')$

$= AB + AD' + A'B'CD' + B'CD' + A'B'C'D' + A'B'C'D$

	00	01	11	10
00	1	1	0	1
01	0	0	0	0
11	1	1	1	1
10	1	0	0	1

$$\text{SOP: } f = \sum (0, 1, 2, 8, 10, 12, 13, 14, 15)$$

$$\text{POS: } F = \pi (3, 4, 5, 6, 7, 9, 11)$$

$$d) F(A, B, C) = (A' + B)(B' + C) \\ = A'B' + A'C + BC$$

A \ BC	00	01	11	10
0	1	1	1	0
1	0	0	1	0

$$\text{SOP: } f = \sum (0, 1, 3, 7)$$

$$\text{POS: } F = \pi (2, 4, 5, 6)$$

$$e) F(x, y, z) = 1$$

x \ yz	00	01	11	10
0	1	1	1	1
1	1	1	1	1

$$\text{SOP: } f = \sum (0, 1, 2, 3, 4, 5, 6, 7)$$

no max term

$$f) F(x, y, z) = (x + y)(y + xz) \\ = xy + xz + yz + xz$$

x \ yz	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$\text{SOP: } f = \sum (3, 5, 6, 7)$$

$$\text{POS: } F = \pi (0, 1, 2, 4)$$

2.14) a) $F(x, y, z) = \sum (1, 3, 7)$

$= F(x, y, z) = \pi (0, 2, 4, 5, 6)$

b) $F(A, B, C, D) = \sum (0, 2, 6, 11, 13, 14)$

$F(A, B, C, D) = \pi (1, 3, 4, 5, 7, 8, 9, 10, 12, 15)$

c) $F(x, y, z) = \pi (0, 3, 6, 7)$

$\Rightarrow F(x, y, z) = \sum (1, 2, 4, 5)$

d) $F(A, B, C, D) = \pi (0, 1, 2, 3, 4, 6, 12)$

$\Rightarrow F(A, B, C, D) = \sum (5, 7, 8, 9, 10, 11, 13, 14, 15)$

2.15) The main difference between canonical and standard form is the canonical form is a way of representing Boolean output of digital circuit using boolean algebra while standard form is simplified function version of canonical form.

2.16) a) $n = 3$

$f(x, y, z) = \sum (0, 1, 2, 3, 4, 5, 6, 7)$

$x \backslash yz$	00	01	11	10
0	1	1	1	1
1	1	1	1	1

$f(x, y, z) = 1$

b) K map

$$+ 2R_x + z'R_x + z'R_x + z'R_x + z'R_x + z'R_x + z'R_x + z'R_x + z'R_x$$

$$xyz = 1$$

2.17) a) $n=3$

2.17) a) $n=3$
 $f(x, y, z) = \pi(0, 1, 2, 3, 4, 5, 6, 7)$

0	0	0	0	0	0
1	0	0	0	0	0

$$f(x, y, z) = 0$$

b) $f(x, y, z) = 1$

$$b) f(x, y, z) = 1$$

when we can use duality principle,

$$(x'+y'+z')(x'+y'+z)(x+y+z)(x'+y+z')$$

$$(x+y+z)(x+y+z)(x+y+z)(x+y+z) = 0$$

2.18) X-OR: $x \oplus y = x'y + x y'$

Dual of X-OR: $(x+y)(x+y)$

$$= \dot{p}_R + \dot{p}_x + \dot{p}_y + \dot{p}_z$$

$$R'_x + R_x =$$

$$= X-NOR$$

complement

complement of XOR (XNOR) = $(x \oplus y)'$
 $= (x'y + xy')$

$$(x+y)(x+y')$$

$$= xx + xy' + yx + yy'$$

$$= xy + xy'$$

2.19 a) Inhibition = $x \cdot y = xy' = x$ not y
 commutative law =

we have to check $xy = yx$

$$xy = xy' \quad yx = yx'$$

$$xy \neq yx$$

Associative property

$$(xy)z = x(yz)$$

$$L.H.S = (xy)z$$

$$= (xy')z$$

$$= xy'z$$

$$R.H.S = x(yz)$$

$$= x(yz')$$

$$= x(yz')'$$

$$= x(y' + z)$$

$$= xy' + xz$$

$$L.H.S \neq R.H.S$$

$$b) \quad x \oplus y = xy' + x'y$$

$$y \oplus x = yx' + y'x$$

$$= xy' + x'y$$

$$x \oplus y = y \oplus x \therefore \text{commutative}$$

we have to check, $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

$$(x \oplus y) \oplus z = (xy' + x'y) \oplus z$$

$$= (xy' + x'y)z' + (xy' + x'y)z$$

$$= (xy'z' + x'yz') + (xyz + x'y'z)$$

$$= xyz + x'yz' + xy'z + x'y'z'$$

$$= xyz + x'yz' + xy'z + x'y'z'$$

$$= \sum (1, 2, 4, 7)$$

$$= x \oplus (y \oplus z)$$

c) NAND : $(xy)' = x' + y'$

associative law : $(x \cdot y)z = x(yz)$

so, NAND is not associative

d) NAND $(xy)' = x' + y'$

NOR $(x + y)' = x'y'$

Distributive law : $x(yz) = (xy)(xz)$

so, NAND & NOR are not distributive

2:20)

x	y	z	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

yz	00	01	11	10
x	0	0	1	0
1	0	1	1	1

$F = xz + yz + xz$

2.21)

x	y	z	$x \oplus y$	$F = A \oplus z$
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	0	1

$$A = x \oplus y$$

$$= xy' + xy'$$

$$F = A \oplus z$$

$$= A'z + Az'$$

$$= A \oplus z$$

$$= x \oplus y \oplus z$$

- 2.22) a) 2-input exclusive OR gates: one 14 pin package can contain 6 of these gates.
- b) 3 input AND gate: one 14 pin package contain 4 of these gates.
- c) 4 input NAND gate: one 14 pin package contain 3 of the gates.
- d) 5 input NOR gate: one 14 pin package contain 3 of these gate.
- e) 8 input NAND gates: one 14 pin package contain 2 of these gate.

2.23) Truth table for a positive logic AND gate:

A	B	$A \text{ AND } B (A \cdot B)$
0	0	0
0	1	0
1	0	0
1	1	1

Truth table of negative OR gate / NOR gate

A	B	$A + B$	$(A + B)'$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

Truth table of negative AND / NAND gate

A	B	AB	$(AB)'$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Truth table OR gate

A	B	$A + B$
0	0	0
0	1	1
1	0	1
1	1	1